Arrival delays and flight cancellations

US reporting carriers, January 2025

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Research questions

This project examines the complete January 2025 reporting-carrier file from the US Bureau of Transportation Statistics (BTS). R is used for the arrival-delay analysis, and DuckDB SQL in DBeaver is used to investigate cancellations for OH.

1. How do arrival delays vary by day and across reporting airlines?
2. Does the difference between UA and WN persist when comparing shared directional routes?
3. How concentrated are OH cancellations in particular days and origin airports, and which cancellation reasons were reported?

The results describe this file and this month. They do not establish an airline's performance across all periods.

Data and preparation

The original file is `On_Time_Reporting_Carrier_On_Time_Performance_(1987_present)_2025_1.csv`. It is stored in the project's `data` folder and contains 539,747 flight records for January 1-31, 2025.

The R and SQL blocks below retain the original analysis code. In HTML, expand **Show original R code** or **Show original SQL** to inspect it. The PDF presents the results. Package installation and `View()` commands remain in the source but are not executed during rendering.

Missing values and date range

The original checks show no missing values in `Cancelled` or `Diverted`. `ArrDelay` has missing values, which are excluded from the arrival-delay analysis. After that filtering step, `na.rm = TRUE` is not needed for calculations using `arrivals$ArrDelay`.

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.03022 0.00000 1.00000
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.00216 0.00000 1.00000
```

```
    Min.  1st Qu.   Median     Mean  3rd Qu.     Max.     NA's
 -87.000  -18.000   -8.000    3.757    7.000 3282.000    17478
```

```
[1] "2025-01-01" "2025-01-31"
```

Cancellation rates use all flight records in the selected group. Arrival-delay rates use only completed, non-diverted flights with a known arrival delay. A delayed arrival means `ArrDelay >= 15`. Negative `ArrDelay` values indicate early arrivals. These denominators are different, so a low arrival-delay rate does not by itself imply a low cancellation rate.

Monthly overview

```
  total_flights cancelled_flights cancelled_pct diverted_flights
1        539747             16312      3.022157             1166
```

There were **16,312 cancellations (3.02%)** and **1,166 diversions** among 539,747 flight records.

```
    Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
-87.000  -18.000   -8.000    3.757    7.000 3282.000
```

```
  flights_n delayed_15_pct mean_arrdelay_min median_arrdelay_min
1    522269       18.78917          3.756614                 -8
```

Among **522,269 eligible arrivals**, **18.79%** were at least 15 minutes late. The mean signed arrival delay was **3.76 minutes**, while the median was **-8 minutes**.

The negative median indicates that at least half of eligible flights arrived eight minutes early or earlier. Long positive delays pull the mean upward. The mean here describes deviation from the schedule across eligible arrivals, including early arrivals; it is not the average delay among delayed flights only.

Daily arrival delays

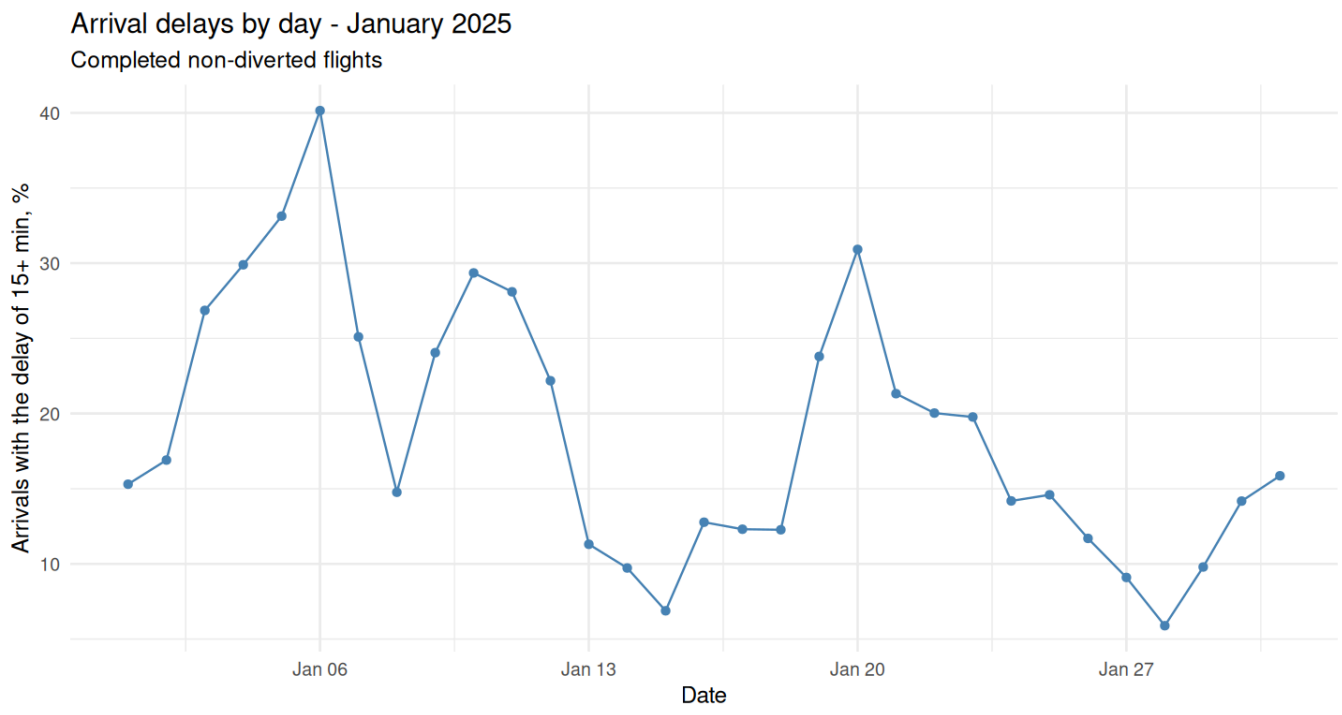


Figure 1: Each daily percentage uses the eligible arrivals on that day.

Arrival-delay distribution

```
50% 75% 90% 95% 99%
 -8   7  37  74 205
```

The 90th percentile was 37 minutes, the 95th percentile was 74 minutes, and the 99th percentile was 205 minutes. These values describe the upper tail of the signed arrival-delay distribution.

```
  early_arrivals_pct delayed_60plus_pct delayed_180plus_pct
1           65.44003           6.365494            1.277694
  mean_delay_among_delayed_min
1                     70.76321
```

65.44% of eligible arrivals were early. **6.37%** were at least 60 minutes late, and **1.28%** were at least 180 minutes late. Among flights delayed by at least 15 minutes, the mean arrival delay was **70.76 minutes**.

Comparing airlines

All reporting airlines

The table combines cancellation rates from all scheduled-flight records with arrival-delay rates from eligible arrivals. Percentages are rounded to two decimal places in the original airline summary code.

Airline	Scheduled flights	Cancelled (%)	Eligible arrivals	Delayed 15+ (%)	Delayed 60+ (%)
F9	15,526	2.52	15,110	25.98	9.99
OH	21,094	9.00	19,151	25.48	10.69
B6	17,918	1.80	17,558	24.46	7.63
G4	9,345	1.27	9,206	23.71	9.25
OO	65,036	2.07	63,502	20.88	8.59
DL	76,306	2.80	74,025	19.32	7.27
AA	75,088	3.73	72,082	19.23	6.75
MQ	21,890	4.71	20,831	18.43	6.55
NK	17,544	3.26	16,946	18.08	5.25
UA	62,007	1.92	60,668	17.63	5.55
AS	18,163	1.34	17,847	17.58	3.92
YX	27,833	4.02	26,683	16.02	4.53
WN	105,307	2.86	102,120	15.21	3.91
HA	6,690	1.91	6,540	13.55	4.19

UA's arrival-delay rate was **17.63%**, compared with **15.21%** for WN, a difference of approximately **2.42 percentage points**. These totals combine different route networks. OH had the highest cancellation rate in this table, **9.00%**, motivating the SQL investigation below.

UA and WN on shared routes

Routes are directional: an origin-destination pair is distinct from the reverse pair. A route is included only if both UA and WN have at least 30 eligible arrivals on it during the month. Each retained route receives equal weight in the route summary.

```
[1] 139
```

Original summary measure	Result
Shared routes	139
Routes with a higher delay rate for UA	64
Routes with a higher delay rate for WN	74
Routes with equal rates	1
UA mean route delay rate	19.45%
WN mean route delay rate	19.60%
Mean gap: UA minus WN	-0.15 percentage points

On the selected shared routes with equal route weights, UA's mean delay rate was about **0.15 percentage points lower** than WN's. This differs from the all-route, flight-weighted comparison. Both the route selection and the weighting change, so this result does not isolate a causal airline effect. The full 139-route table is included in the appendix.

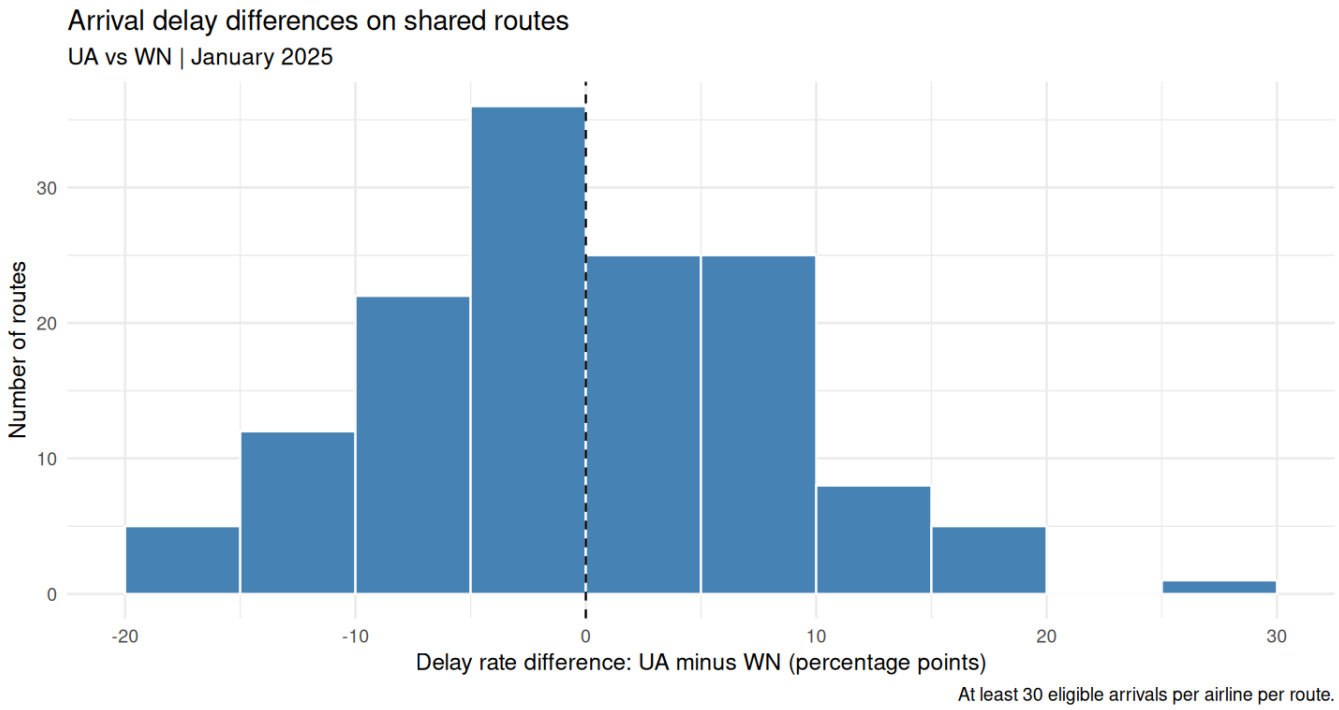


Figure 2: Positive differences mean a higher delay rate for UA. Each shared route contributes one observation.

Sensitivity to the minimum number of flights

```
  min_flights_per_airline shared_routes mean_gap_pp
1                      30           139  -0.1469238
2                      60            90  -0.8858205
3                     100            42  -1.4567722
```

The average UA-minus-WN gap remains negative at the 30-, 60-, and 100-flight thresholds. Its magnitude changes as the retained route set becomes smaller. The direction in this check therefore does not depend exclusively on choosing 30 flights; the check does not establish statistical significance.

Daily comparison on the selected shared routes

The following comparison uses the same monthly set of 139 shared routes. Each daily rate is calculated from the eligible flights for that airline and day. It is flight-weighted within each day, whereas the preceding monthly route summary gives every route equal weight.

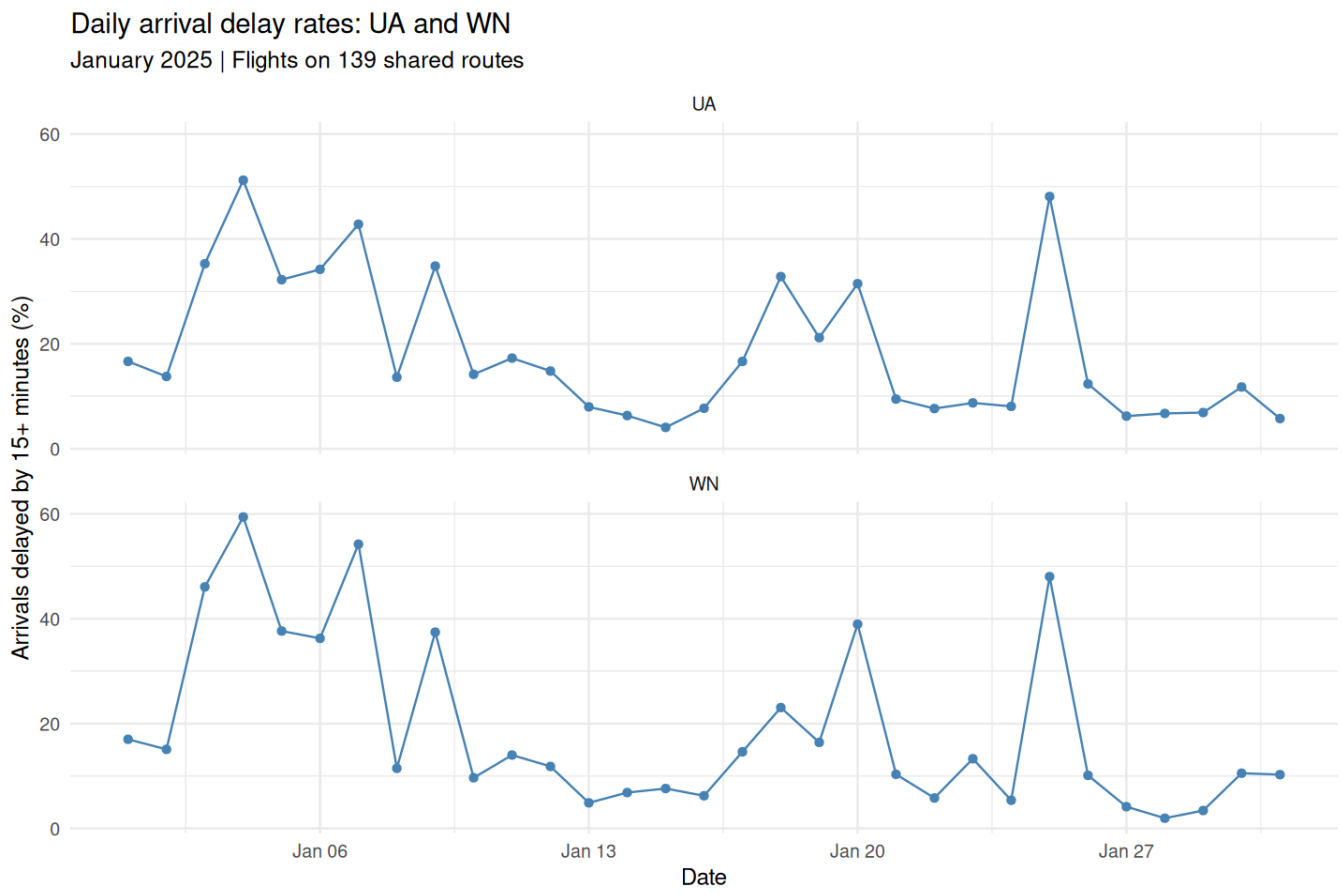


Figure 3: The two panels use the same vertical scale. Daily route mixes and numbers of flights may differ between the airlines.

SQL investigation: OH cancellations

These are the original DuckDB queries used in DBeaver. Their results are recorded as ordinary tables in this report; Quarto does not execute the SQL blocks or open the DBeaver database. The import statement retains the original local file path.

total_flights
539,747

first_date	last_date
2025-01-01	2025-01-31

total_flights	cancelled_flights	cancelled_pct
539,747	16,312	3.02

The SQL total and cancellation rate agree with the R monthly overview.

Daily cancellation rates

The query orders OH's days by cancellation rate. The three highest rates occurred on **January 30, January 10, and January 6**. All 31 daily results are included here.

Date	Scheduled flights	Cancelled flights	Cancelled (%)
2025-01-30	724	314	43.37
2025-01-10	713	262	36.75
2025-01-06	706	235	33.29
2025-01-31	714	215	30.11
2025-01-22	672	170	25.30
2025-01-11	590	91	15.42
2025-01-21	652	99	15.18
2025-01-09	724	102	14.09
2025-01-23	724	92	12.71
2025-01-19	662	80	12.08
2025-01-07	651	64	9.83
2025-01-05	703	62	8.82
2025-01-20	712	35	4.92
2025-01-24	713	30	4.21
2025-01-29	671	25	3.73
2025-01-12	696	11	1.58
2025-01-04	650	4	0.62
2025-01-25	590	3	0.51
2025-01-08	672	3	0.45
2025-01-15	672	2	0.30
2025-01-16	724	0	0.00
2025-01-18	580	0	0.00
2025-01-17	713	0	0.00
2025-01-03	707	0	0.00
2025-01-14	652	0	0.00
2025-01-26	696	0	0.00
2025-01-27	712	0	0.00
2025-01-28	652	0	0.00
2025-01-13	712	0	0.00
2025-01-02	706	0	0.00
2025-01-01	629	0	0.00

Three peak days compared with the rest of the month

Day group	Scheduled flights	Cancelled flights	Cancelled (%)
Three peak days	2,143	811	37.84
Other days	18,951	1,088	5.74

The three selected days accounted for **811 of OH's 1,899 monthly cancellations (42.71%)** and **10.16%** of its scheduled flights. Their combined cancellation rate was **37.84%**, compared with **5.74%** on the remaining days.

These dates were selected after inspecting the daily rates. The comparison describes concentration in this month; it is not a test using dates chosen in advance.

Origin airports during the three peak days

DCA had the largest number of cancelled OH departures on the selected dates: **259 of 366 flights (70.77%)**. CLT followed with **130 of 493 (26.37%)**. The query ranks airports by the number of cancellations, not by the cancellation percentage. The complete 93-origin result is in the appendix.

Reported cancellation reasons

Date	Reported reason	Cancelled flights
2025-01-06	Weather	234
2025-01-06	Carrier	1
2025-01-10	Weather	259
2025-01-10	Carrier	3
2025-01-30	Carrier	301
2025-01-30	Weather	13

On January 6, **234 of 235** cancellations were classified as Weather. On January 10, **259 of 262** were classified as Weather. On January 30, **301 of 314** were classified as Carrier.

The reported categories differ across the selected dates. They do not identify the specific events behind each cancellation, and a carrier classification alone does not establish a particular operational cause.

Conclusions and limitations

1. Cancellations affected 3.02% of flight records. Among eligible arrivals, 18.79% were at least 15 minutes late. Early arrivals were common, while long delays pulled the mean above the median.
2. The UA-WN comparison depends on the route set and the weighting. UA's rate was higher in the all-route comparison, but slightly lower when giving each selected shared route equal weight. The shared-route gap remained negative across the three minimum-flight thresholds.
3. OH cancellations were concentrated in three days. Weather dominated the reported cancellations on January 6 and 10, while Carrier dominated on January 30.

This is a descriptive analysis of one month. It does not control for departure times, weather exposure, aircraft, or other operating conditions. The arrival-delay comparison excludes cancellations and diversions, which must be considered separately. Small differences should not be treated as proof that one airline is inherently more reliable.

Reproducing the report

Keep `AviationReport.qmd` in the project folder and the original CSV in its `data` subfolder. Open the project in RStudio, install the packages using the existing installation lines if needed, and render the document as HTML. The HTML can also be printed to PDF.

The original R commands are preserved in their original order. Only Quarto chunk options control whether installation commands, `View()` windows, or large console outputs run or appear during rendering. The three figures are generated by the original plotting code. No extra R analysis or SQL queries are included.

The formatted airline and route tables, SQL tables, and written interpretations are the checked **January 2025** results stored in the document. They do not update automatically if a different dataset is substituted. The executing R chunks and figures do recalculate. SQL is run separately in DBeaver; its import path must match the location of the original CSV on the computer running it. The original `CREATE TABLE` statement is for the initial import into a database without an existing `flights` table.

Sources

- BTS January 2025 source archive.
- BTS field definitions.
- BTS reporting instructions and cancellation codes.

Appendix: complete detailed results

The following tables contain all rows from the original route comparison and peak-day origin-airport query. Route rates and gaps are displayed to two decimal places; the original R calculations use unrounded route rates.

All 139 shared directional routes

Origin	Destination	UA arrivals	UA delayed (%)	WN arrivals	WN delayed (%)	Gap (pp)
JAX	DEN	37	35.14	30	10.00	25.14
DEN	PSP	67	29.85	40	10.00	19.85
MIA	DEN	95	24.21	32	6.25	17.96
MEM	DEN	60	30.00	31	12.90	17.10
HDN	DEN	89	39.33	43	23.26	16.07
DEN	ORD	258	31.01	101	15.84	15.17
MKE	DEN	31	25.81	96	11.46	14.35
DEN	HDN	88	37.50	43	23.26	14.24
ORD	LAS	176	16.48	31	3.23	13.25
DEN	MKE	31	35.48	94	23.40	12.08
HNL	LAX	73	15.07	31	3.23	11.84
ORD	AUS	80	27.50	31	16.13	11.37
DEN	SNA	122	29.51	118	18.64	10.86
BOI	DEN	87	20.69	91	9.89	10.80
ORD	DEN	262	17.56	101	7.92	9.64
LAX	PHX	86	13.95	160	5.00	8.95
BNA	ORD	56	16.07	54	7.41	8.66
MCO	DEN	147	26.53	160	18.12	8.41
BZN	DEN	120	31.67	73	23.29	8.38
IND	DEN	61	31.15	96	22.92	8.23
ELP	DEN	55	18.18	60	10.00	8.18
ORD	BWI	63	19.05	54	11.11	7.94
AUS	ORD	80	17.50	31	9.68	7.82
DEN	LAS	223	31.84	259	24.32	7.51
DEN	LGA	162	27.16	116	19.83	7.33
RNO	DEN	102	22.55	92	15.22	7.33
BWI	DEN	65	21.54	133	14.29	7.25
IAD	DEN	190	21.58	62	14.52	7.06
LAX	HNL	71	7.04	30	0.00	7.04

Origin	Destination	UA arrivals	UA delayed (%)	WN arrivals	WN delayed (%)	Gap (pp)
DEN	JAX	37	27.03	30	20.00	7.03
DEN	COS	75	28.00	95	21.05	6.95
DEN	RNO	102	28.43	93	21.51	6.93
DTW	DEN	62	19.35	40	12.50	6.85
BWI	ORD	63	22.22	51	15.69	6.54
ABQ	DEN	62	16.13	102	9.80	6.33
COS	DEN	75	20.00	95	13.68	6.32
PHL	DEN	68	19.12	61	13.11	6.00
TUS	DEN	79	11.39	99	6.06	5.33
SNA	DEN	124	19.35	121	14.05	5.31
OMA	DEN	117	21.37	122	16.39	4.97
LAS	DEN	223	18.83	259	13.90	4.93
DEN	ABQ	62	24.19	103	19.42	4.78
LAX	BWI	34	11.76	38	7.89	3.87
RDU	DEN	62	25.81	59	22.03	3.77
MCI	DEN	116	18.97	149	15.44	3.53
DEN	SAT	99	28.28	104	25.00	3.28
STL	DEN	88	19.32	149	16.11	3.21
GEG	DEN	68	8.82	64	6.25	2.57
BUR	DEN	62	17.74	98	15.31	2.44
TPA	DEN	123	18.70	98	16.33	2.37
MSY	DEN	100	21.00	89	19.10	1.90
DEN	ONT	63	23.81	95	22.11	1.70
BOS	DEN	140	13.57	67	11.94	1.63
GRR	DEN	62	30.65	31	29.03	1.61
CLE	DEN	68	25.00	34	23.53	1.47
DEN	MCI	116	21.55	153	20.26	1.29
DEN	TUS	80	22.50	99	21.21	1.29
SAN	DEN	173	16.18	187	14.97	1.21
ONT	DEN	64	15.62	97	14.43	1.19
DEN	PHL	68	25.00	62	24.19	0.81
SAT	DEN	100	17.00	104	16.35	0.65
DEN	LAX	272	19.85	159	19.50	0.36
DEN	MIA	96	22.92	31	22.58	0.34
IAD	PHX	54	25.93	31	25.81	0.12

Origin	Destination	UA arrivals	UA delayed (%)	WN arrivals	WN delayed (%)	Gap (pp)
DEN	DSM	62	19.35	31	19.35	0.00
DEN	OMA	117	24.79	121	24.79	-0.01
LAX	DEN	272	11.40	155	11.61	-0.22
CMH	DEN	62	27.42	65	27.69	-0.27
MSP	DEN	117	17.95	128	18.75	-0.80
LAX	LAS	151	15.89	203	16.75	-0.85
BNA	DEN	101	13.86	163	14.72	-0.86
PSP	DEN	67	11.94	39	12.82	-0.88
DEN	AUS	142	23.24	174	24.14	-0.90
DEN	IAD	191	19.90	62	20.97	-1.07
ORD	BNA	55	12.73	50	14.00	-1.27
DEN	TPA	123	21.95	99	23.23	-1.28
DEN	SAN	170	19.41	187	20.86	-1.44
SLC	DEN	142	19.01	215	20.47	-1.45
ATL	DEN	79	17.72	93	19.35	-1.63
DEN	GEG	68	27.94	64	29.69	-1.75
DEN	PHX	252	17.86	313	20.13	-2.27
PHX	DEN	251	14.34	319	16.61	-2.27
DEN	BUR	63	20.63	96	22.92	-2.28
DEN	FLL	125	24.00	34	26.47	-2.47
BWI	LAX	34	11.76	35	14.29	-2.52
SFO	LAX	274	10.22	101	12.87	-2.65
DEN	BOI	87	24.14	89	26.97	-2.83
SEA	DEN	147	8.84	94	11.70	-2.86
DEN	SFO	301	21.26	124	24.19	-2.93
AUS	DEN	142	11.97	174	14.94	-2.97
DEN	BOS	140	17.14	68	20.59	-3.45
MTJ	DEN	92	20.65	41	24.39	-3.74
PHX	IAD	55	9.09	31	12.90	-3.81
PIT	DEN	62	24.19	39	28.21	-4.01
DEN	DTW	62	25.81	40	30.00	-4.19
LGA	DEN	163	14.72	116	18.97	-4.24
SJC	DEN	92	9.78	127	14.17	-4.39
DEN	MCO	147	19.73	161	24.22	-4.50
DEN	SJC	96	12.50	127	17.32	-4.82

Origin	Destination	UA arrivals	UA delayed (%)	WN arrivals	WN delayed (%)	Gap (pp)
DEN	SDF	49	18.37	30	23.33	-4.97
DEN	SLC	142	20.42	215	25.58	-5.16
PHX	SFO	122	7.38	94	12.77	-5.39
LAX	SFO	276	11.23	102	16.67	-5.43
DEN	MEM	60	26.67	31	32.26	-5.59
SAN	SFO	184	12.50	121	18.18	-5.68
DEN	CVG	59	13.56	31	19.35	-5.80
DEN	BWI	64	18.75	133	24.81	-6.06
DEN	MSP	118	19.49	125	25.60	-6.11
SFO	SAN	182	10.99	122	17.21	-6.22
DEN	GRR	62	19.35	31	25.81	-6.45
DSM	DEN	62	16.13	31	22.58	-6.45
DEN	ATL	80	17.50	96	23.96	-6.46
DEN	BNA	99	18.18	163	25.15	-6.97
DEN	BZN	121	30.58	71	38.03	-7.45
DEN	ELP	56	17.86	63	25.40	-7.54
DEN	SMF	126	17.46	140	25.00	-7.54
SFO	DEN	300	12.67	124	20.97	-8.30
ORD	PHX	164	10.37	32	18.75	-8.38
PHX	LAX	85	5.88	160	14.37	-8.49
DEN	SEA	146	19.86	94	28.72	-8.86
DEN	IND	61	16.39	96	26.04	-9.65
LAS	LAX	151	13.25	200	23.00	-9.75
DEN	PDX	147	23.13	108	33.33	-10.20
PDX	DEN	146	10.96	107	21.50	-10.54
DEN	MTJ	92	20.65	41	31.71	-11.06
DEN	STL	89	16.85	149	28.86	-12.01
SFO	PHX	121	9.09	94	21.28	-12.19
FLL	DEN	125	19.20	35	31.43	-12.23
LAS	SFO	201	12.94	107	25.23	-12.30
DEN	CLE	68	20.59	33	33.33	-12.75
SMF	DEN	127	8.66	138	21.74	-13.08
SFO	LAS	201	13.43	107	27.10	-13.67
DEN	RDU	62	16.13	59	30.51	-14.38
DEN	MSY	101	12.87	91	27.47	-14.60

Origin	Destination	UA arrivals	UA delayed (%)	WN arrivals	WN delayed (%)	Gap (pp)
CVG	DEN	59	13.56	31	29.03	-15.47
LAS	ORD	176	9.66	31	25.81	-16.15
DEN	PIT	62	19.35	39	35.90	-16.54
DEN	CMH	62	22.58	66	39.39	-16.81
PHX	ORD	167	9.58	32	28.12	-18.54

All 93 OH origin airports on the three peak days

Origin	Scheduled flights	Cancelled flights	Cancelled (%)
DCA	366	259	70.77
CLT	493	130	26.37
DFW	114	20	17.54
PHL	104	19	18.27
ATL	30	17	56.67
HSV	26	13	50.00
BNA	25	13	52.00
BHM	26	12	46.15
HPN	24	11	45.83
TYS	28	10	35.71
SAV	30	10	33.33
CVG	25	10	40.00
CAE	19	10	52.63
CHS	23	9	39.13
SDF	13	9	69.23
GSP	32	9	28.13
ORF	28	9	32.14
PNS	22	8	36.36
DAY	31	8	25.81
CLE	14	8	57.14
BUF	11	8	72.73
MHT	17	7	41.18
GSO	19	6	31.58
ALB	10	6	60.00
BGR	9	6	66.67
ILM	16	6	37.50
CHA	16	6	37.50

Origin	Scheduled flights	Cancelled flights	Cancelled (%)
SYR	11	6	54.55
BTV	17	6	35.29
LEX	15	6	40.00
TUL	14	6	42.86
PWM	14	6	42.86
PVD	11	6	54.55
BDL	13	6	46.15
AVL	15	6	40.00
SHV	22	6	27.27
CRW	11	6	54.55
MYR	23	5	21.74
MSY	7	5	71.43
PIT	13	5	38.46
IAD	13	5	38.46
DTW	14	5	35.71
MEM	13	5	38.46
LIT	15	4	26.67
MSN	11	4	36.36
ROC	8	4	50.00
MSP	10	4	40.00
DSM	7	3	42.86
CMH	15	3	20.00
GRR	4	3	75.00
ICT	10	3	30.00
MLB	12	3	25.00
MKE	7	3	42.86
SBN	9	3	33.33
CAK	15	3	20.00
GPT	14	3	21.43
JAX	7	3	42.86
XNA	3	2	66.67
OKC	4	2	50.00
AGS	8	2	25.00
MCI	5	2	40.00
RDU	7	2	28.57
DAB	12	2	16.67

Origin	Scheduled flights	Cancelled flights	Cancelled (%)
JAN	19	2	10.53
CID	4	2	50.00
STL	6	2	33.33
VPS	11	2	18.18
TLH	12	2	16.67
MGM	8	2	25.00
RIC	4	1	25.00
IND	5	1	20.00
ABE	9	1	11.11
BTR	9	1	11.11
OMA	3	1	33.33
LAN	2	1	50.00
MLI	6	1	16.67
PIA	9	1	11.11
OAJ	7	1	14.29
RSW	2	1	50.00
AVP	3	1	33.33
ECP	6	1	16.67
IAH	1	0	0.00
SRQ	4	0	0.00
LFT	8	0	0.00
BWI	2	0	0.00
FWA	6	0	0.00
HOU	2	0	0.00
MOB	2	0	0.00
FAY	1	0	0.00
PBI	1	0	0.00
SGF	5	0	0.00
TRI	1	0	0.00
GNV	5	0	0.00